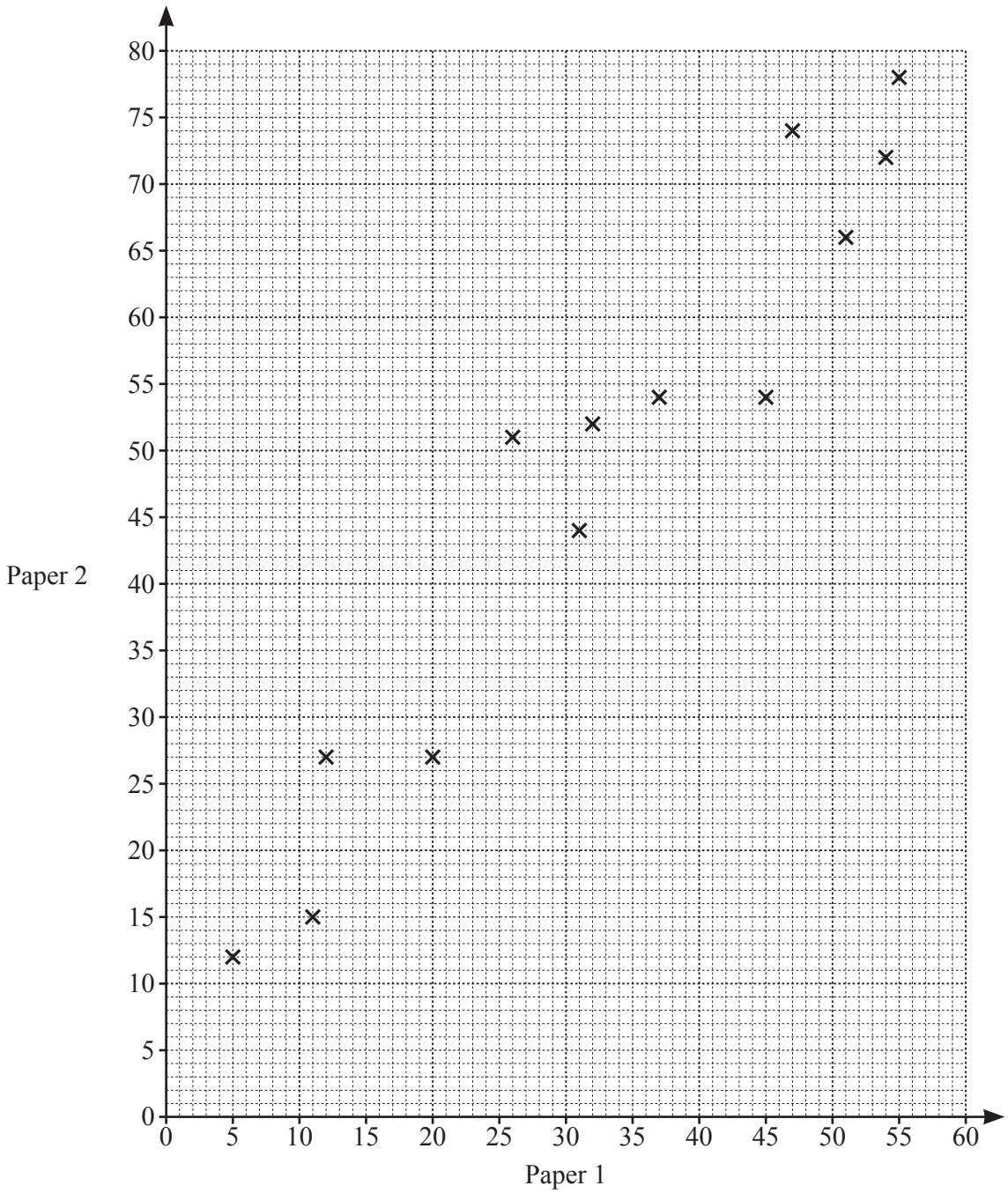
0580_s22_qp_31_Q7

7 **(a)** A class of 15 students take two tests in science, paper 1 and paper 2.
The scores for each student are shown in the table.

Paper 1	5	11	12	20	26	31	32	37	45	47	51	54	55	23	42
Paper 2	12	15	27	27	51	44	52	54	54	74	66	72	78	30	58



(i) Complete the scatter diagram.
The first thirteen points have been plotted for you. [1]

(ii) What type of correlation is shown in the scatter diagram?

..... [1]

(iii) On the grid, draw a line of best fit. [1]

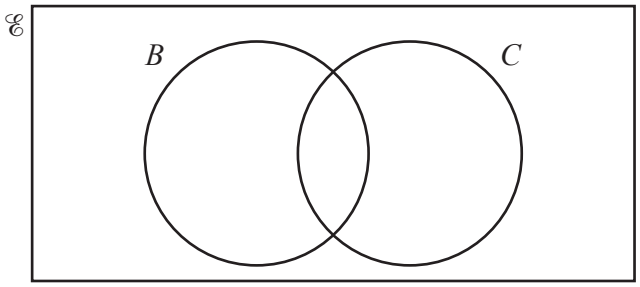
(iv) Another student scores 24 on paper 1.

Use your line of best fit to find an estimate for their score on paper 2.

..... [1]

(b) 140 students choose which subjects they want to study.

- 122 students choose biology (B).
- 55 students choose chemistry (C).
- 2 students do not choose biology and do not choose chemistry.



(i) Complete the Venn diagram.

[2]

(ii) One of these students is picked at random.

Find the probability that this student chooses biology and chemistry.

..... [1]